CUSTOMER SEGMENTATION PROJECT

Using RFM Model & K-Means Clustering

Case Study: Hospitality Business

Author	Year	Method
Farhan Fadhilah Rasyid	**2024**	**RFM + K-Means**

Project Overview

Business Problem	The business had no data-driven understanding of their customer types, making it difficult to design targeted marketing and loyalty programs.
Solution	Applied RFM (Recency, Frequency, Monetary) scoring combined with K-Means Clustering to segment customers into actionable groups.
Dataset	183 customers, 110 days of transaction data (Feb – May, 2024)
Result	3 distinct customer segments identified with tailored marketing strategies recommended for each.
Tools Used	Python · Pandas · Scikit-learn · Matplotlib · Yellowbrick

Methodology

Step 1 — Data Collection

Collected 110 days of customer booking data from the business booking records covering February to May, 2024. Fields included Customer ID, Check-in/Check-out dates, room type, stay type, and price.

Customer ID	Check-in dates	Check-out dates	Room type	Stay type	Price
001AND	01/02/2024	29/02/2024	*Superior*	Bulanan	2700000
002DEA	01/02/2024	29/02/2024	*Superior*	Bulanan	2700000
0016RID	21/02/2024	26/02/2024	*Standard Double*	Harian	200000
0018AHM	28/02/2024	28/02/2024	*Standard Double*	Harian	200000
…	…	…	…	…	…

Step 2 — Data Preprocessing

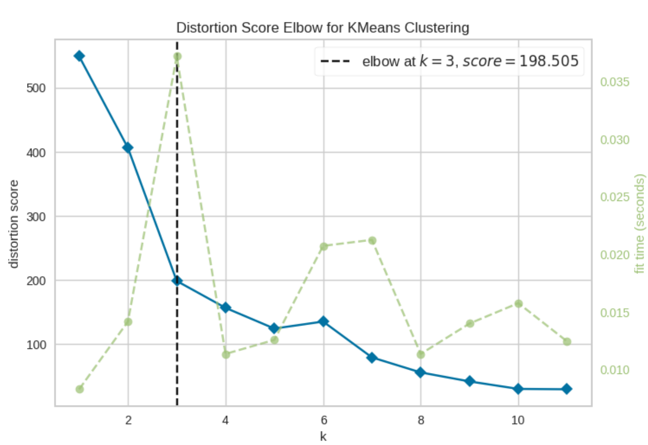
- Data Selection: Transformed raw transaction records into an RFM-ready dataset. Higher scores on Frequency and Monetary indicate more valuable customers. Lower Recency scores indicate more recent (and thus more engaged) customers.
  Customer ID: ID_Pelanggan,
  Recency: Hari_Sejak_Pemesanan_Terakhir,
  Frequency: Jumlah_Pemesanan,
  Monetary: Jumlah_Harga.
- Data Cleansing: Removed duplicates and handled null values — none were found in the final dataset.

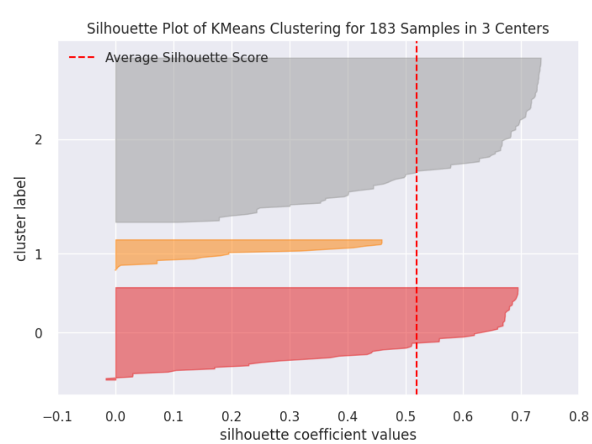
ID_Pelanggan	Hari_Sejak_Pemesanan_Terakhir	Jumlah_Pemesanan	Jumlah_Harga
001AND	52	2	5400000
002DEA	52	2	5400000
0016RID	86	1	200000
0018AHM	84	2	400000
…	…	…	…

Metric	Stands For	What It Measures	High Value Means
R	Recency	Days since last transaction	Customer purchased very recently
F	Frequency	Number of transactions made	Customer purchases often → loyal
M	Monetary	Total spending amount	Customer contributes high revenue

Step 3 — K-Means Clustering

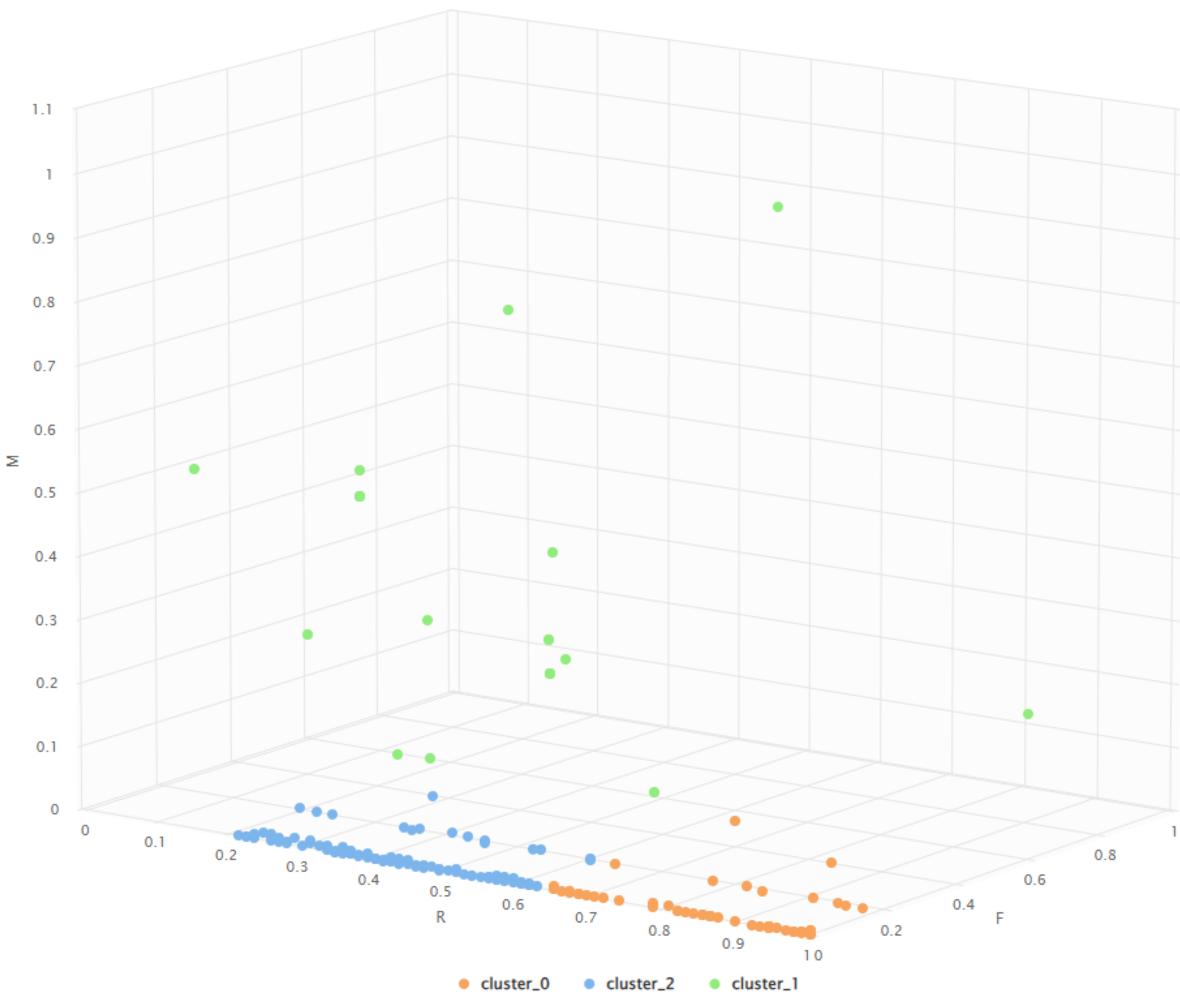
- Used the Elbow Method to determine optimal number of clusters.
- The elbow point appeared at k = 3 (distortion score = 198.505), confirming 3 clusters.
- Applied K-Means with k = 3 on normalized RFM values.
- Validated cluster quality using Silhouette Coefficient = 0.521 (good separation).





Clustering Results

The 183 customers in the dataset were segmented into three distinct groups based on their RFM profiles:



Segment	Cluster	# Customers	Recency	Frequency	Monetary
Loyal / Active	1	20	0.426	0.300	0.376
New Customers	2	106	0.424	0.022	0.005
Inactive	0	57	0.860	0.025	0.007

Note: R, F, M values are normalized (0–1). Lower R = more recent. Higher F & M = more engaged and higher-spending.

Recommended Marketing Strategies

Based on the cluster characteristics, the following strategies are recommended for each customer segment:

Loyal / Active	• Implement advanced loyalty program with VIP tiers • Offer exclusive deals and early access perks • Use cross-sell & up-sell with premium services • Invite for product feedback and surveys
New Customers	• Send welcome email with first-time discount • Introduce loyalty program early to drive retention • Offer promotions to incentivize repeat bookings • Personalized onboarding experience
Inactive	• Re-engagement campaign via email/SMS/social media • Offer exclusive win-back discount or promo • Send satisfaction survey to understand churn reasons • Showcase new services since their last visit

Key Takeaways

- The majority of customers (106 / 58%) are new customers with low engagement — there is a large opportunity to convert them into loyal customers.
- Only 20 customers (11%) are truly loyal — they represent the most valuable segment and should be prioritized for retention.
- 57 inactive customers (31%) represent potential re-engagement revenue if the right campaigns are deployed.
- A Silhouette score of 0.521 confirms the clustering model produced well-separated, meaningful segments.
- Future improvements: Add customer satisfaction scores, integrate real-time segmentation, or explore alternative algorithms (e.g., DBSCAN, Hierarchical Clustering).

Skills & Tools Used

Data Analysis	Machine Learning	Visualization
Excel	Scikit-learn	Matplotlib
Pandas	K-Means	Seaborn
NumPy	Rapid Miner	Yellowbrick